

Week 8: Random Signals

Homework, Weekly Quiz, Complete Solutions, and Instructor Grading Notes

Course context: DSP + Machine Learning for a mixed senior / first-year ECE cohort. DSP reference: Oppenheim & Schaffer. ML reference: Raschka et al.

Correlation convention used in this packet: $r_{xy}[m] = E\{X[n]Y^*[n+m]\}$. For real signals and $y = h*x$, this convention gives $r_{xy} = r_{xx}^*h$. The commonly seen reversed-h expression corresponds to the alternate convention $E\{X[n+m]Y^*[n]\}$.

Learning targets

- Explain ensemble, realization, stationarity, WSS, and the role of ergodicity.
- Interpret mean, variance, autocorrelation, cross-correlation, and correlation matrices.
- Propagate mean and second-order statistics through an LTI system.
- Analyze white noise and explain noise coloring.
- Connect correlation and random-signal models to ML preprocessing and augmentation.

Homework — Student Version

Problem 1 — Ensemble, realization, and WSS (10 pts) [Core]

A temperature sensor is sampled once per minute. You repeat the same 30-minute experiment on 100 nominally identical days. Let $X[n]$ denote the resulting random process.

- What is the ensemble? What is one realization?
- Suppose $E\{X[n]\} = 22^\circ\text{C}$ and $r_{xx}[n_1, n_2] = 484 + 2 \exp(-0.1|n_2 - n_1|)$. Is $X[n]$ WSS? Explain.
- Does WSS alone guarantee that the average of one 30-minute record equals 22°C ? Explain.

Problem 2 — A random-amplitude sinusoid (12 pts) [Core / Senior derivation]

Let $X[n] = A \cos(\omega_0 n + \Theta)$, where A is a fixed positive constant and Θ is uniformly distributed on $[0, 2\pi)$. Assume Θ is independent of n .

- Find $\mu_x[n]$.
- Find $r_{xx}[m]$.
- Is the process WSS? What feature of the random phase makes the second-order statistics independent of absolute time?

Problem 3 — Correlation matrix properties (12 pts) [Core]

A zero-mean WSS process has normalized autocorrelation $r_{xx}[m] = \rho^{|m|}$ with $\rho = 0.6$.

- Construct the 4×4 correlation matrix for $[X[0], X[1], X[2], X[3]]^T$, assuming $r_{xx}[0] = 1$.
- State two structural properties of the matrix that follow from WSS.
- Explain, without computing every eigenvalue, why the matrix must be positive semidefinite.

Problem 4 — Template matching and alignment (12 pts) [Core + ML]

A known length-25 template $s[n]$ occurs once in a noisy length-200 observation $x[n]$, but at an unknown delay. You compute $c[k] = \sum_n x[n+k]s[n]$. The largest value occurs at $k = 73$.

- (a) What does $k = 73$ mean under this definition?
- (b) Give one reason correlation-based alignment can improve a downstream classifier.
- (c) Give one case where raw correlation peak magnitude can be misleading, and suggest a remedy.

Problem 5 — Mean through an LTI system (10 pts) [Core]

A WSS input has mean $\mu_x = 2.5$. Find the output mean for each filter.

- (a) $h_1[n] = \{1/4, 1/2, 1/4\}$
- (b) $h_2[n] = \{1, -1\}$
- (c) Explain the result of part (b) in terms of DC gain.

Problem 6 — White noise through an FIR filter (14 pts) [Core]

Zero-mean white noise $X[n]$ has variance $\sigma_x^2 = 9$ and drives the FIR filter $h[n] = \{1/2, 1/2\}$.

- (a) Find μ_Y .
- (b) Find σ_Y^2 .
- (c) Find $r_{YY}[0]$ and $r_{YY}[1]$.
- (d) Is $Y[n]$ white? Explain physically why neighboring output samples are correlated.

Problem 7 — Input-output correlation convention (14 pts) [Senior derivation]

Use the convention $r_{xy}[m] = E\{X[n]Y[n+m]\}$ for real-valued signals. Let $Y[n] = \sum_k h[k]X[n-k]$.

- (a) Starting from the definition, derive $r_{xy}[m]$ in terms of $r_{xx}[m]$ and $h[k]$.
- (b) State the corresponding formula for $r_{yy}[m]$.
- (c) A classmate writes $r_{xy} = r_{xx} * h[-m]$. Is that necessarily wrong? Explain carefully.

Problem 8 — Designing realistic ML noise augmentation (16 pts) [DSP + ML synthesis]

You are training a classifier for vibration signals from a rotating machine. Measurements show that environmental noise is stronger at low frequencies and exhibits significant correlation for about 8 samples. Design a training-time augmentation strategy. Your answer should specify:

- (a) how to generate the noise,
- (b) how to control its severity,
- (c) where in the train/validation/test workflow to apply it,
- (d) one experiment that tests whether the augmentation actually improves robustness,
- (e) why independent white Gaussian noise alone may be a poor model for this application.

Homework — Complete Solutions

Solution 1 — Ensemble, realization, and WSS (10 pts) [Core]

Solution

- The ensemble is the collection of all possible 30-minute temperature sequences generated by the experiment. Each day's 30-minute sequence is one realization (sample path).
- The mean is constant: $E\{X[n]\} = 22$. The autocorrelation depends on n_1 and n_2 only through the lag $n_2 - n_1$ because $|n_2 - n_1|$ is shift-invariant. The second moment is finite. Therefore the process is WSS.
- No. WSS does not imply that a time average from one finite realization equals the ensemble expectation. That requires an ergodicity assumption plus enough data for a useful estimate.

Grading guidance:

- 3 pts ensemble/realization distinction; 4 pts WSS reasoning; 3 pts stationarity vs ergodicity.

Solution 2 — A random-amplitude sinusoid (12 pts) [Core]

Solution

$$X[n] = A \cos(\omega_0 n + \Theta), \quad \Theta \sim \text{Uniform}[0, 2\pi)$$

(a) Averaging cosine over a uniformly random phase gives zero:

$$\mu_x[n] = E\{X[n]\} = 0$$

(b) Use $\cos \alpha \cos \beta = (1/2)[\cos(\alpha - \beta) + \cos(\alpha + \beta)]$. The term involving 2Θ averages to zero.

$$r_{xx}[m] = E\{X[n]X[n+m]\} = (A^2/2) \cos(\omega_0 m)$$

(c) The mean is constant and the autocorrelation depends only on m , so the process is WSS. The uniformly random phase removes dependence on the absolute time index n in the ensemble average.

Grading guidance:

- 3 pts mean; 6 pts autocorrelation derivation/result; 3 pts WSS interpretation.

Solution 3 — Correlation matrix properties (12 pts) [Core]

Solution

$$R = \begin{bmatrix} 1 & .6 & .36 & .216 \\ .6 & 1 & .6 & .36 \\ .36 & .6 & 1 & .6 \\ .216 & .36 & .6 & 1 \end{bmatrix}$$

- The matrix is symmetric (Hermitian in the complex case).
- It is Toeplitz: every diagonal is constant because entries depend only on lag.
- It is positive semidefinite because for any vector a , $a^H R a = E\{|a^H x|^2\} \geq 0$. An expected squared magnitude cannot be negative.

Grading guidance:

- 5 pts matrix; 3 pts structural properties; 4 pts PSD explanation.

Solution 4 — Template matching and alignment (12 pts) [Core]

Solution

- Under $c[k] = \sum x[n+k]s[n]$, the peak at $k = 73$ means the template best matches the observation beginning at approximately sample 73.
- Alignment removes irrelevant timing variability. Features are then extracted in a common coordinate system, so the classifier can focus on shape/amplitude/frequency differences rather than arbitrary delay.
- Raw correlation magnitude depends on signal/template energy. A high-energy but poor-shape segment can produce a large score. Normalize by local signal energy and template energy (normalized cross-correlation), or standardize amplitudes before matching.

Grading guidance:

- 4 pts lag interpretation; 4 pts ML benefit; 4 pts failure mode + remedy.

Solution 5 — Mean through an LTI system (10 pts) [Core]

Solution

$$\mu_y = \mu_x \sum_k h[k]$$

(a) $\sum h_1 = 1$, so $\mu_y = 2.5$.

(b) $\sum h_2 = 0$, so $\mu_y = 0$.

(c) h_2 is a first-difference / highpass filter with zero DC gain. A constant mean is a DC component, so it is rejected.

Grading guidance:

- 3 pts each numerical result; 4 pts DC-gain explanation.

Solution 6 — White noise through an FIR filter (14 pts) [Core]

Solution

(a) The input is zero mean, so $\mu_y = 0$.

$$\sigma_y^2 = \sigma_x^2 \sum |h[k]|^2 = 9[(1/2)^2 + (1/2)^2] = 9/2 = 4.5$$

(c) For white input, $r_{yy}[m] = \sigma_x^2 r_{hh}[m]$. For $h = \{1/2, 1/2\}$:

$$r_{hh}[0] = 1/2, \quad r_{hh}[1] = 1/4$$

$$r_{yy}[0] = 4.5, \quad r_{yy}[1] = 2.25$$

(d) $Y[n]$ is not white because $r_{yy}[1] \neq 0$. Consecutive outputs share one input sample: $y[n] = (x[n] + x[n-1])/2$ and $y[n+1] = (x[n+1] + x[n])/2$.

Grading guidance:

- 2 pts mean; 4 pts variance; 4 pts correlation values; 4 pts whiteness/physical explanation.

Solution 7 — Input-output correlation convention (14 pts) [Core]

Solution

Start with the definition and substitute the LTI relation:

$$\begin{aligned}r_{xy}[m] &= E\{X[n]Y[n+m]\} = E\{X[n] \sum_k h[k]X[n+m-k]\} \\r_{xy}[m] &= \sum_k h[k] r_{xx}[m-k] = (r_{xx} * h)[m]\end{aligned}$$

For the output autocorrelation (real h):

$$r_{yy}[m] = r_{xx}[m] * h[m] * h[-m] = r_{xx}[m] * r_{hh}[m]$$

The classmate is not necessarily wrong; the expression with $h[-m]$ is associated with the alternate definition $\tilde{r}_{xy}[m] = E\{X[n+m]Y[n]\}$. Cross-correlation lag conventions differ, so the definition must be stated before deciding whether a reversal is required.

Grading guidance:

- 6 pts derivation; 4 pts output autocorrelation; 4 pts convention explanation.

Solution 8 — Designing realistic ML noise augmentation (16 pts) [Core]

Solution

- Generate white noise and pass it through a stable lowpass or measured/estimated noise-shaping filter whose autocorrelation lasts roughly 8 samples. An FIR kernel estimated from real background-noise recordings is a particularly defensible choice.
- Scale the resulting noise to a randomly selected SNR (for example, draw SNR uniformly from a range representing mild through severe deployment conditions). Normalize the generated colored noise before SNR scaling so severity is controlled separately from spectral shape.
- Apply stochastic augmentation only to training examples. Keep validation and ordinary test data unaugmented so metrics remain interpretable; create additional fixed stress-test sets with known noise conditions.
- Compare a baseline model, white-noise augmentation, and matched colored-noise augmentation on clean data plus several fixed colored-noise SNRs. Report both clean accuracy and robustness degradation.
- Independent white Gaussian noise has zero autocorrelation at nonzero lags and flat PSD. The measured nuisance is low-frequency dominated and temporally correlated, so white-only augmentation does not reproduce the deployment distribution.

Grading guidance:

- 3 pts noise model; 3 pts severity/SNR; 3 pts workflow; 4 pts experimental design; 3 pts why white noise is inadequate.

Weekly Quiz — Student Version

Suggested time: 15–20 minutes. Total: 20 points. Students may use a basic calculator but no notes.

1. (2 pts) In one sentence, distinguish an ensemble from a realization.

2. (2 pts) State the two main second-order conditions for WSS.

3. (2 pts) True or false: Every white random process must have a Gaussian amplitude distribution. Briefly justify.

4. (3 pts) A WSS process has $r_{xx}[m] = 5(0.7)^{|m|}$ and $\mu_x = 0$. What are the variance and $r_{xx}[2]$?

5. (3 pts) A WSS input has mean 4 and passes through $h[n] = \{0.25, 0.5, 0.25\}$. Find the output mean.

6. (3 pts) Zero-mean white noise has variance 2 and drives $h[n] = \{1, -1\}$. Find the output variance.

7. (2 pts) Why does filtering white noise generally make the output colored?

8. (3 pts) Give one reason correlation-based alignment can help an ML classifier, and one reason noise augmentation should model realistic correlation/spectral structure.

Weekly Quiz — Solutions and Grading

1. Ensemble vs realization

The ensemble is the set of possible sample paths from the stochastic model; one realization is one particular observed sequence.

Grading: 2 pts: both parts clearly distinguished.

2. WSS conditions

The mean is constant in time, and the autocorrelation depends only on lag (with finite second moment).

Grading: 1 pt each main condition; accept equivalent wording.

3. White vs Gaussian

False. White describes second-order correlation / flat PSD. Gaussian describes the amplitude/joint distribution. Gaussian white noise is a special case.

Grading: 1 pt false; 1 pt explanation.

4. Read statistics from r_{xx}

Because $\mu_x = 0$, variance = $r_{xx}[0] = 5$. Also $r_{xx}[2] = 5(0.7)^2 = 2.45$.

Grading: 1.5 pts each.

5. Output mean

$\sum h = 1$, so $\mu_y = 4$.

Grading: 1 pt sum; 2 pts answer/interpretation.

6. Output variance

$\sigma_y^2 = 2(1^2 + (-1)^2) = 4$.

Grading: Full credit for correct white-noise variance formula and result.

7. Why noise becomes colored

Filtering causes neighboring outputs to share input samples and produces nonzero output autocorrelation at nonzero lags; equivalently, $|H(e^{j\omega})|^2$ shapes a flat input PSD.

Grading: 2 pts for either clear time- or frequency-domain explanation.

8. ML synthesis

Alignment removes nuisance delay so fixed features correspond to the same event location. Realistic augmentation should mimic deployment nuisance statistics; otherwise the model may train for the wrong robustness problem. Full credit: 1.5 pts for each idea.

Instructor Notes and Differentiation

For first-year students

- Prioritize physical interpretations: correlation = similarity across delay; WSS = stable first/second-order behavior; filtering white noise creates memory.
- For Problem 7, allow students to follow the derivation line-by-line rather than derive from scratch.
- Use simulations to make ensemble and expectation concrete before introducing matrix notation.

For senior students

- Require explicit correlation convention and correct reversal/conjugation handling.
- Ask for covariance-matrix form $C_Y = H C_X H^H$ as an extension.
- Add a PSD extension: $S_{YY}(e^{j\omega}) = S_{XX}(e^{j\omega})|H(e^{j\omega})|^2$ and connect to spectral factorization / whitening.

Common misconceptions to watch for

- Stationarity and ergodicity are not synonyms.
- White and Gaussian are not synonyms.
- Autocorrelation at lag zero is average power, not always variance when the mean is nonzero.
- Cross-correlation reversal depends on the lag convention.
- Noise augmentation is not automatically beneficial; unrealistic nuisance distributions can create distribution mismatch.